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Spatio-Temporal Graph Neural Networks Integrating Weather Features and Uncertainty Quantification for Urban Rail Transit Ridership Prediction in Malaysia

ASSIGNMENT 1

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Abstract

An accurate and effective urban rail transit ridership prediction system provides multi-fronts benefits to many, including users, governments, and the environment. Traditional approaches leveraging statistical methods and machine learning techniques capture only temporal dependencies which disregard the critical spatial relationship that is intrinsically present between urban rail transit stations. Recently, Graph Neural Networks (GNN) has seen an increase in application attributed to the fact that it has the capacity to discern both temporal and spatial dependencies concurrently. Nonetheless, existing literature presents several limitations. Firstly, there is insufficient inclusion of external factors such as weather conditions and unplanned events. Next, limited attention is given to uncertainty quantification which ridership and travel demands inherently contains a certain degree of randomness. Thirdly, a predominant quantity of literature leverages datasets that are based in China and America which have temperate climates, indicating a gap in validating such approaches on Southeast Asia's tropical climate. Hence, this research proposes a spatio-temporal GNN framework integrating weather features and uncertainty quantification for urban rail transit ridership prediction. The research aims to develop a robust GNN-based prediction model accustomed to Malaysian urban rail transit context, to evaluate the impact of weather features and uncertainty quantification on prediction accuracy. The expected contribution includes a STGNN architecture adapted for Malaysian context with empirical evidence on the influence of weather features and uncertainty quantification on ridership in tropical climates.

1 Introduction (Research Background)

In recent years, public transportation ridership has seen a surge worldwide due to its various benefits for users, society and the environment. Looking at the user perspective, public transport allows users to save cost and avoid traffic congestion. As the number of private cars on the road are reduced, greenhouse gases emissions are subsequently reduced as well, providing another benefit from an environmental standpoint [1]. As public transport becomes increasingly relied upon for both economic and environmental reasons, operators face growing pressure to maintain consistent service quality, optimise operations and meet rising demand. With respect to abovementioned reasons, understanding ridership patterns alongside accurate ridership prediction approaches have undoubtedly become crucial.

An accurate ridership forecast enables a more optimised resources allocation across multiple dimensions, including optimal operational scheduling, routing, staff capacity planning and fare adjustments. Consequently, these optimisations cascade to secondary benefits such as reduced electrical consumption, lower maintenance costs and an elevated passenger satisfaction. Taken together, these outcomes empower respective authorities to make informed, data-driven decisions that consistently enhance both operational efficiency and overall service quality.

Prediction of ridership is not a novel application as various methodologies have been proposed over the years. Initially, it started with classical time series methods like ARIMA (Autoregressive Integrated Moving Average), then it advances to machine learning techniques like XGBoost, followed by deep learning methods including Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) and their variants. While these approaches are able to capture temporal patterns, their limitation lies in the fact that fundamentally they capture only temporal

dimensions of ridership data. As a result, important spatial dependencies that influence ridership prediction remain unmodelled, which restricts the overall predictive performance.

Rail ridership data would ideally contain essential information about stations, volume of passengers, and timestamps. This structure of this data is multi-dimensional which encompasses both temporal insights (ridership by time) and spatial insights (distance between stations). Graph Neural Networks (GNN) has lately emerged as an effective approach with capability in capturing both of these dimensions, also known as spatio-temporal dimension. In essence, the original tabular format of the ridership data would be converted as graphs, where stations are represented as nodes while the connections between stations are the edges. This transformation enables GNNs to model how ridership at one station influences neighbouring stations, which is a critical consideration that conventional models are unable to represent.

Despite this strong potential, existing literature often presents GNNs superficially without fully explaining the reasons for outperforming traditional and deep learning techniques. Fundamentally, GNNs are modelled to learn relational structure, meaning they do not only capture temporal patterns but also propagate information across interconnected nodes. This enables GNNs to uncover latent patterns such as passenger flow propagation, inter-station influence and network-level anomalies [2].

Building upon this, Spatio-Temporal Graph Neural Networks (STGNN) extend on the GNN framework by modelling both spatial and temporal concurrently. While GNNs focus primarily on relational structure, STGNNs integrate additional temporal learning mechanisms such as recurrent layers, temporal convolutions or attention modules to capture how ridership evolves across time. This dual capacity allows STGNNs to represent the connectivity of rail stations and the sequential behavior of passenger flow, making them particularly suitable for urban rail ridership prediction [2].

Scoping on the Malaysian context, the development of urban rail transit has expedited with the introduction of the expansion of Light Rail Transit 3 (LRT3) in 2025 and Mass Rapid Transit 3 (MRT3) network in 2032. In parallel, Malaysia has an inherent nature of having a tropical climate of relatively consistent weather throughout the year with distinct patterns. This presents a unique consideration for ridership prediction modelling. However, it has not been adequately addressed in existing literature, which largely focus more on temperate climate regions. Therefore, Malaysia offers an opportunity to explore whether spatio-temporal modelling behaves differently under tropical climate conditions. It also allows an investigation of whether GNN-based models are able to enhance reliability in situations where climate variability is low.

2 Research Problem Statement

Although significant progress and outcome has been realised with regards to transit ridership prediction, several research gaps are still present. These gaps might deter the reliability of existing prediction systems, particularly in the context of Malaysia’s urban rail transit.

Gap 1: Limited Type of Features

Existing GNN models for ridership prediction predominantly utilise historical ridership data, temporal features and spatial relationship. Liu et al. [2] and Li et al. [3] achieved excellent results using MultiDimensional Temporal Dependency Graph Neural Network (MTDGNN) and Dynamic Spatio-Temporal Graph Fusion Network (DSTGFN) respectively. However, both studies acknowledged and mentioned the limitation of leveraging additional features such as weather conditions or unforeseen circumstances like incidents or large-scale events where ridership volume rises or declines unexpectedly. Although Farahmand et al. [4] investigated interplay of weather conditions towards bus ridership prediction, their research did not include spatial features nor was it tested on urban rail transit. This gap is especially important to address for tropical climates like Malaysia where weather patterns differ from the temperate regions in most of the existing literature.

Gap 2: Lack of Uncertainty Inclusion

The majority of literature produces deterministic point estimates of demand which in return ignores the inherent uncertainty in predictions. As addressed by the research of Wang et al. [5], the lack of uncertainty element during the prediction process hinders the potential in a robust real-time prediction. With that, their proposed Probabilistic Graph Neural Networks (Prob-GNN) demonstrated that including uncertainty elements was successful in improving performance of models. However, it is noted that a combination of uncertainty quantification with weather features might not have been extensively explored, posing a potential limitation towards the application of probabilistic prediction in operations that are weather-sensitive.

Gap 3: Limited Geographical and Climate Context

A large quantity of literature utilised metro system dataset originated from China, including but not limited to, cities such as Beijing, Shanghai, Hangzhou and Chongqing. Aside from China, American transit networks (Chicago and New York) are also present, while Zeb et al. [6] conducted their research using Rawalpindi–Islamabad Metro transit data in Pakistan. To the best of the author's knowledge, there has yet to be any research utilising GNN-based ridership prediction models for Southeast Asian context, including Malaysia. Malaysia inherently possesses a tropical climate with year-round consistent temperature and humidity as well as its own distinctive cultural ridership patterns. This uniqueness implies a possibility that different geographical contexts, the likes of Malaysia, might result in varying model performance and feature importance compared to previously studied regions.

Thus, this research aims to address these gaps by proposing a framework that integrates weather features and uncertainty quantification in a Spatio-Temporal Graph Neural Networks (STGNN) architecture for urban rail transit ridership prediction in Malaysian context. The purpose is to improve the reliability of urban rail ridership prediction in the Malaysian tropical climate context.

3 Research Questions

Anchoring on the identified research gaps, this study intends to address the following research questions:

RQ1: Amongst weather, uncertainty quantification, spatial and temporal features, which has the strongest predictive power and feature importance for urban rail ridership prediction in tropical climates such as Malaysia?

RQ2: How can a Spatio-Temporal Graph Neural Network (STGNN) be modelled to effectively integrate the selected features for accurate ridership prediction?

RQ3: How does the proposed STGNN perform compared to the baseline model when evaluated across standard prediction metrics?

4 Research Objectives

Following the SMART (Specific, Measurable, Achievable, Relevant, Time-bound) approach, this research has such objectives in mind:

RO1: To experiment and analyse the predictive power and feature importance of weather, uncertainty quantification, spatial and temporal features for urban rail transit ridership prediction in a tropical climate context.

RO2: To design and develop a Spatio-Temporal Graph Neural Network (STGNN) architecture that effectively integrates the selected features.

RO3: To evaluate the performance of the proposed STGNN against baseline models (ARIMA, XGBoost, LSTM) across standard evaluation metrics such as MAE, RMSE and MAPE.

5 Research Scope and Contribution

Ridership prediction can be classified as either short-term or long-term prediction, of which this research focuses on short-term ridership prediction of 10 to 60 minutes intervals. The research will limit itself only to the urban rail transit system of KTM Komuter in Malaysia. As for the spatial scope, it encompasses station-level ridership where relationships between stations will be modeled through graph structure. For temporal scope, it includes hourly, daily and weekly patterns, for both regular and special event periods such as holidays and major public events.

That said, long-term ridership prediction beyond one day is not included. Furthermore, the research will not be extended for other transportation means such as bus, taxi, LRT or MRT.

The research intends to make contributions through theoretical and practical perspective as follows:

Theoretical contribution:

1. Development of a STGNN architecture integrating weather features uniquely designed for tropical climate contexts, filling the geographical research gap of Southeast Asia's climate.
2. Establish a methodological framework that is adaptable and reproducible for other tropical countries' transit systems to generalize the research findings beyond Malaysian context.

3. Integration of uncertainty quantification with weather features in one singular framework as opposed to fragmented analysis.

Practical contribution:

1. Provide decision-makers data-driven insights of not only point predictions, but also uncertainty intervals for a more optimised and robust resources allocation and contingency planning.
2. Validation of STGNN-based ridership prediction on Malaysia's urban rail transit data to provide local benchmark and insights.

6 Critical Analysis

Author(s) & Year	Research Focus	Focus Type	Contribution	Model(s)	Methodology	Dataset (Country)	Prediction	Evaluation Metrics	Limitation
Li et al. (2025)	Multi-dimensional time-dependent dynamic graph neural network	Architectural Innovation	Captures both short-term and **long-term temporal features**	GCN, GRU, GAT	1D dilated convolutions to extract multi-dimensional temporal dependencies	Southwest China and Central China Metro (China)	Metro Ridership	MAE,RMSE, Accuracy	Computational costly with 70 million rows; lack of external factors (weather, special-event, incidents)
Liu et al. (2025)	Dynamic Spatio-Temporal Graph Fusion Network modeling	Architectural Innovation	Develop non-negative tensor decomposition techniques for **dynamic virtual graphs**	TCN, GCN, GAT	Parallel learning architecture; Distributed processing for reducing computational costs	Shanghai and Hangzhou Metro (China)	Metro Ridership	MAE, RMSE, MAPE, R^2	Lack of additional features (weather, special-event, incidents); high model complexity
Zeb et al. (2025)	Forecasting public transit ridership amidst COVID-19	Feature Integration	Consideration of **unprecedented circumstance of COVID-19**; integration of pandemic-related features	LSTM	Deep learning with seasonality, COVID-19 cases and non-pharmaceutical interventions as features	Rawalpindi Islamabad Metro (Pakistan)	BRT Ridership	MAE, RMSE, MAPE	Performance deteriorates for long-term prediction due to uncertainty; limited to specific pandemic context
Ma et al. (2025)	Spatio-temporal traffic flow forecasting based on second-order continuous graph neural network	Architectural Innovation	Addresses over-smoothing in deep GNNs; **second-order continuous framework** with ILN layers	SCGNN, ILN	Second-order continuous framework for higher-order spatio-temporal dependencies; multiple SCGNN layers integrated	California and Los Angeles traffic data (USA)	Traffic Flow	MAE, RMSE, MAPE	Not suitable for real-time due to time complexity and large model size; limited explainability; high computational cost
Wang et al. (2024)	Uncertainty quantification of spatiotemporal travel demand with probabilistic graph neural networks	Feature Integration	Quantify **spatiotemporal uncertainty** in travel demand; develop probabilistic framework	GCN, GAT, LSTM	Probabilistic assumptions; MLE training with NLL loss; multi-graph proximity	Chicago rail, bus and ridesharing data (USA)	Metro, Bus and Rideshare Demand	NLL, Calibration Error, MPIW, PICP, MAE, MAPE	Requires further validation on other context and cities; limited to specific probabilistic distributions

Wu et al. (2025)	LLM4PT: A large language model-based system for public transit prediction	Architectural Innovation	**LLM for transit prediction**; provides human-readable explanations	LLaMA3.1-8B	Prompt tuning with Chain-of-Thought Spatiotemporal Aligner with learnable parameters	Hangzhou, Nanjing, New York metro data (China, USA)	Metro OD Flow	WAPE, MAPE	Limited public event data; fine-tuning LLMs remains complex
Zou et al. (2024)	Real-time prediction during underground incidents	Feature Integration	Consideration of **incident feature** in prediction	ConvLSTM	Incident matrices combined with external features	Beijing AFC metro incident data (China)	Metro OD Flow	MMSE, MMAE, MMAPE	Limited to incident scenarios; requires real-time incident data; may not generalize to other disruption types
Ren et al. (2025)	A multimodal deep learning-based direct ridership model for planning-oriented demand prediction	Feature Integration	Integrates **land-use images with 5D built environment indicators**	CNN	Land-use image generation from multi-source data; image-based SHAP for interpretability	Beijing URT Network (China)	Metro OD Ridership	MAE, RMSE, MAPE, R^2	Limited to planning scenarios; requires high-quality spatial data
Zhao et al. (2022)	Short-term metro ridership prediction during unplanned events	Feature Integration	Addresses prediction under **unplanned events**	LSTM, Random Forest, gradient-boosting decision trees	SMOTE to handle minority data of unplanned events	Large-scale Metro AFC data (Undisclosed)	Metro Ridership	MAPE, RMSE	Limited to specific unplanned event types
Farahmand et al. (2023)	Predicting bus ridership based on the weather conditions using deep learning algorithms	Feature Integration	Research **impact of weather** on bus ridership	MLP	MLP with meteorological parameters (temperature, precipitation, wind, humidity); hourly aggregation	Twente Region Bus Network data (Netherlands)	Bus Ridership	MAE, EVS	Limited to short-term prediction; weather forecast uncertainty;

Table 1: Comparative Analysis Table

As depicted in Table 1, several trends can be observed through systematic analysis and organisation of the literature.

Observation 1: Methodological Limitations and Progressions

The first observation is related to the evolution of ridership prediction techniques which commenced with statistical techniques like ARIMA, extending to machine learning approaches, subsequently proceeding with deep learning methods, and lastly GNNs.

Prediction of ridership can be tagged as a time series problem, and hence classical time series approach such as ARIMA was utilised in the early days. These statistical methods are able to capture temporal information, patterns and seasonality. One of the limitations lies in the fact that these statistical methods often assume linear relationships [7]. To address the non-linearity gap, machine learning techniques with the likes of Support Vector Machines (SVM), Random Forest (RF) Gradient Boosting Decision Trees (GBDT) are deployed. Zhao et al. [8] compared RF and GBDT with deep learning approaches for unplanned events and indicated that a tree-based method is able to handle imbalance data with the proper oversampling technique. However, such machine learning models require manual feature engineering and meticulous feature selection which can be complex as it requires a heightened understanding of the data and domain knowledge. Next, deep learning techniques were introduced and immediately garnered attention and wide application attributed to its elevated capabilities in discerning information. Convolutional Neural Networks (CNN) is better at capturing spatial dependencies while Recurrent Neural Networks (RNN) and its variants such as LSTM excel at capturing temporal dependencies. Zeb et al. [6] demonstrated LSTM's capabilities through predicting transit ridership during unprecedented COVID-19 circumstances that integrates seasonal data, pandemic statistics and non-pharmaceutical interventions as features. Despite the compelling performance result, these deep neural network approaches generally focus on only one dimension, and have limitations in capturing the interplay between both. To overcome all the previously stated shortcomings, GNN was developed. GNN is able to model urban rail transit networks as a graph, where nodes represent the stations while the connections between the stations are represented as edges. This way, GNN is able to discern jointly the spatio-temporal dependencies without needing to learn them one after another in a sequential manner. Li et al. [3], Liu et al. [2], Ma et al. [7] all leveraged variants of spatio-temporal GNNs in ridership prediction and obtain results that bypass baseline models, clearly signifying GNN's strength for use-cases that are spatial and temporal dependent.

Observation 2: Integration of External Features and Uncertainty Quantification

One common gap observed across the literature is the lack of integration of external features, particularly weather and unforeseen event features.

Farahmand et al. [4] carried out a research on the impact of weather conditions towards the bus ridership prediction using MLP technique. Specifically, the MLP utilised same-hour and next-hour weather conditions combined with additional features such as large events, holidays and bus cancellations. The findings indicated that same-hour conditions perform well during regular weather but fall short during adverse weather conditions. Contextually, this can be linked to the authors' geographically data located in the Netherlands which represents a temperate climate with higher temperature variance. This can be further extended and explored in tropical climates like Malaysia. A comparison between the weather influence on different climates would further strengthen the influence weather has on ridership prediction.

Wang et al. [5] addressed a key gap by introducing probabilistic forecasting through their uncertainty quantification framework for spatiotemporal travel demand. Their approach utilises Probabilistic Graph Neural Networks (PGNN), generating probabilistic prediction intervals rather than deterministic point estimates. The research demonstrated that uncertainty quantification is particularly valuable during high-variance periods and extreme events, where traditional deterministic models fail to capture inherent variability.

However, the integration of weather features within uncertainty quantification frameworks remains underexplored. While Wang et al. [5] established the importance of probabilistic predictions, their framework did not specifically examine how weather variability contributes to prediction uncertainty.

Observation 3: Limitation of Geographical and Climate Context

Analysis of the 10 datasets across the reviewed literature reveals a significant tendency towards certain geographical contexts. It is observed that 5 were based in China, 3 were in the USA, 1 in the Netherlands while 1 in Pakistan. China and the USA, which are fundamentally countries with temperate climates, tend to dominate the dataset, implying that the majority of the results were validated on climate and weather with pronounced seasonal fluctuations. Li et al. [3], Liu et al. [2], Wu et al. [9], Zou et al. [10], Ren et al. [11] all utilised dataset from big metropolitan cities in China including Beijing, Shanghai, Hangzhou and Nanjing. For the American counterpart, Ma et al. [7], Wang et al. [5], Wu et al. [9] experimented their approach on main metropolitan cities like Chicago, New York and Los Angeles.

Shifting the attention to Malaysia, or even Southeast Asia as a whole region that represents a tropical climate which exhibits minimal seasonal variation, with temperature remaining relatively stable throughout the year. Furthermore, the large Chinese and American cities used as the dataset all surpass the population of Malaysia's capital. Thus, the geographical context difference here extends beyond mere comparison between temperate and tropical climate, it also includes the population count, urban infrastructure environment, socioeconomic factors and distinct cultural mobility patterns tied to regional festival celebrations. With such comparable difference in terms of geographical context, validating STGNN-based ridership prediction models on Malaysian context would provide paramount insights into model generalisability across diverse geographical contexts.

7 Summary

Based on the comparative analysis table and extended elaboration above, two principal research gaps are identified:

1. **Integration of weather features with uncertainty quantification:** Limitation in terms of integration of both weather features with uncertainty quantification in a singular GNN architecture framework. While Wang et al.[5] quantified uncertainty while other literature utilised weather features, these two remains separated.
2. **Tropical climate validation:** Majority of weather impact studies focus solely on temperate regions with high seasonal variance, overlooking the validation of STGNN-based ridership prediction on tropical climates that have annual-long constant weather conditions.

These two critical gaps collectively justify the proposed research direction, to develop a STGNN framework integrating weather and uncertainty quantification specifically validated in Malaysian tropical climate context.

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